



First Name: _____

Family Name: _____

Student Number: _____

Diploma of Engineering
University Entrance Program
Foundation Studies Program

DPEN022 – Enabling Mathematics B

Integration Exam: Sample

This exam represents 35% of the total marks for the subject

Time allowed:	1 hour and 20 minutes
Items permitted in exam:	UOW approved calculator
Aids supplied for exam:	Formula Sheet

INSTRUCTIONS TO STUDENTS

1. Write in blue or black pens only. Pencils are permitted for diagrams and graphs.
2. All questions are to be attempted.
3. All notation is as given in lectures.
4. Show all necessary WORKING and reasoning in the spaces provided.
5. You must achieve a minimum mark of 40% in the Integration Exam. If this requirement is not met, your final mark for the subject will be a Technical Fail.
6. Total number of questions: 16
 - Questions 1-6 are multiple choice.
 - Each question is worth 1 mark.
 - Marks are awarded for the answer only.
 - Circle the letter (A, B, C, D, or E) corresponding to the correct answer of each question.
 - Questions 7-16 are long answer questions.
 - The value of each question is indicated.
 - Part marks may be awarded.

No examination material may be removed from the examination venue.

Question 1 [Primitive Functions]

What is the best first step to evaluate the primitive function of $\frac{dy}{dx} = \left(x - \frac{3}{x}\right)^2$.

$$\left(x - \frac{3}{x}\right)^2 = \left(x - \frac{3}{x}\right) \times \left(x - \frac{3}{x}\right) = x^2 - 6 + \frac{9}{x^2}$$

Now you can integrate $\left(x^2 - 6 + \frac{9}{x^2}\right)$ i.e. $\int x^2 - 6 + \frac{9}{x^2} dx$

OPTION B**Question 2 [Integrals of Power of x]**

Find $\int \sqrt{x} \left(2 + \frac{1}{\sqrt{x}}\right) dx$.

$$\int \sqrt{x} \left(2 + \frac{1}{\sqrt{x}}\right) dx = \int (2\sqrt{x} + 1) dx$$

$$= \frac{2x^{\frac{3}{2}}}{\frac{3}{2}} + x + c$$

$$= \frac{4\sqrt{x^3}}{3} + x + c$$

OPTION A**Question 3 [Definite Integrals of Power of x]**

Evaluate $\int_{-1}^2 (2x^3 + 2) dx$

$$\int_{-1}^2 (2x^3 + 2) dx = \left[\frac{x^4}{2} + 2x \right]_{-1}^2$$

$$= \left[\left(\frac{2^4}{2} + 2 \times 2 \right) - \left(\frac{(-1)^4}{2} + 2 \times (-1) \right) \right]$$

$$= \left[12 - \left(\frac{-3}{2} \right) \right] = \frac{27}{2}$$

OPTION E**Question 4 [Area]**

The area bounded by the graph of $f(x) = x^3 - 4x^2 - x + 4$ and the x -axis can be calculated by:

The graph crosses the x -axis at $x = -1, 1, 4$, so the area is split into two regions.

The function is above the x -axis on $[-1, 1]$ and below the x -axis on $[1, 4]$, so the second integral must have an absolute value sign.

OPTION C**Question 5 [Integration of exponential/logarithmic or trigonometric functions]**

Find $\int 3\cos\left(\frac{x}{3}\right) dx$

$$\int 3\cos\left(\frac{x}{3}\right) dx = 3 \int \cos\left(\frac{x}{3}\right) dx$$

$$= 3 \times 3 \int \frac{1}{3} \cos\left(\frac{x}{3}\right) dx = 9 \sin\left(\frac{x}{3}\right) + c$$

OPTION A**Question 6 [Integration by Simple Substitution]**

The substitution that can be used to find $\int (x-3)(x^2 - 6x + 2)^{3/2} dx$ is:

OPTION E

Question 7 (2 marks) [Applications of the Primitive Function]

The gradient function of a curve is $\frac{dy}{dx} = 1 + 3\sqrt{x}$. If the curve passes through the point (4, 10), find the equation of the curve.

$$\frac{dy}{dx} = 1 + 3\sqrt{x} = 1 + 3x^{\frac{1}{2}}$$

$$y = \int 1 + 3x^{\frac{1}{2}} dx$$

$$y = x + \frac{3x^{\frac{3}{2}}}{\frac{3}{2}} + c \Rightarrow y = x + 2x^{\frac{3}{2}} + c$$

Find c by subbing in (4,10)

$$10 = 4 + 2 \times 4^{\frac{3}{2}} + c$$

$$10 = 4 + 16 + c$$

$$c = -10$$

$$\therefore y = x + 2x^{\frac{3}{2}} - 10$$

Question 8 (2 marks) [Integration of Power of x]

Find the indefinite integral of $\frac{x^5 + 9}{3x^2}$, writing your answer in positive index form.

$$\begin{aligned} \int \frac{x^5 + 9}{3x^2} dx &= \int \frac{x^5}{3x^2} + \frac{9}{3x^2} dx \\ &= \int \frac{x^3}{3} + 3x^{-2} dx \\ &= \frac{x^4}{12} + \frac{3x^{-1}}{-1} + c = \frac{x^4}{12} - \frac{3}{x} + c \end{aligned}$$

Question 9 (3 marks) [Integration of Power of x]

Find $\int \sqrt{x^3} - \frac{4}{x^3} dx$, writing your answer in surd and positive index form.

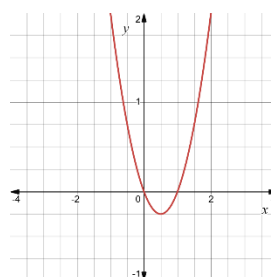
$$\begin{aligned} \int \sqrt{x^3} - \frac{4}{x^3} dx &= \int x^{\frac{3}{2}} - 4x^{-3} dx \\ &= \frac{x^{\frac{3}{2}+1}}{\frac{3}{2}+1} - 4 \times \frac{x^{-3+1}}{-3+1} + c \\ &= \frac{2x^{\frac{5}{2}}}{5} - 4 \times \frac{x^{-2}}{-2} + c = \frac{2x^{\frac{5}{2}}}{5} + \frac{2}{x^2} + c \text{ positive index form} \\ &= \frac{2\sqrt{x^5}}{5} + \frac{2}{x^2} + c \text{ surd form} \end{aligned}$$

Question 10 (3 marks) [Area]

Find the area of the region bounded between the parabola $y = x^2 - x$ and the x -axis from $x = 0$ and $x = 1$. Hint: You need graph the parabola.

Notice that the area is under the x -axis between $x = 0$ and $x = 1$, therefore we must take the absolute value of the definite integral.

$$\begin{aligned} \left| \int_0^1 (x^2 - x) dx \right| &= \left| \left[\frac{x^3}{3} - \frac{x^2}{2} \right]_0^1 \right| \\ &= \left| \left(\frac{1^3}{3} - \frac{1^2}{2} \right) - \left(\frac{0^3}{3} - \frac{0^2}{2} \right) \right| \\ &= \left| -\frac{1}{6} \right| = \frac{1}{6} \text{ units}^2 \end{aligned}$$



Question 11 (4 marks) [Area]

Use algebra to show that the points of intersections of graphs $y = x^2$ and $y = 2 - x^2$ are $(-1, 1)$ and $(1, 1)$ and calculate the area of the region bounded by the graphs $y = x^2$ and $y = 2 - x^2$ shown in the graph below.

$$y = x^2 \quad (1)$$

$$y = 2 - x^2 \quad (2)$$

(1) into (2)

$$x^2 = 2 - x^2 \Rightarrow 2x^2 - 2 = 0$$

$$2x^2 = 2 \Rightarrow x^2 = 1$$

$$x = \pm 1$$

$$\text{when } x = -1, y = (-1)^2 = 1$$

$$\text{when } x = 1, y = (1)^2 = 1$$

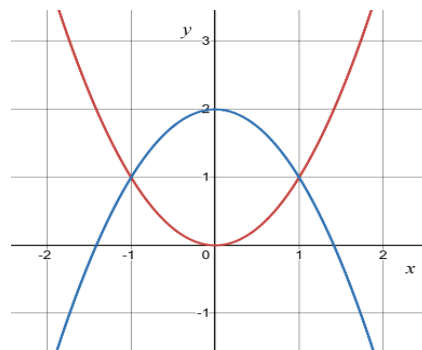
$$\int_{-1}^1 2 - x^2 - x^2 dx = \int_{-1}^1 2 - 2x^2 dx$$

$$= \left[2x - \frac{2x^3}{3} \right]_{-1}^1$$

$$= \left(2 \times 1 - \frac{2 \times 1^3}{3} \right) - \left(2 \times (-1) - \frac{2 \times (-1)^3}{3} \right)$$

$$= \left(2 - \frac{2}{3} \right) - \left(-2 + \frac{2}{3} \right)$$

$$= \frac{8}{3} \text{ units}^2$$

**Question 12 (3 marks) [Integration of exponential or logarithmic or trigonometric functions]**

Find $\int x^2 + e^{\frac{x}{4}+2} dx$.

$$\int x^2 + e^{\frac{x}{4}+2} dx = \int x^2 dx + \int e^{\frac{x}{4}+2} dx$$

$$= \frac{x^3}{3} + 4 \int \frac{1}{4} e^{\frac{x}{4}+2} dx$$

$$= \frac{x^3}{3} + 4e^{\frac{x}{4}+2} + c$$

Question 13 (2 marks) [Integration of exponential or logarithmic or trigonometric functions]

Find $\int \frac{x}{x^2 + 5} dx$.

$$\int \frac{x}{x^2 + 5} dx$$

$$= \frac{1}{2} \int \frac{2x}{x^2 + 5} dx$$

$$= \frac{1}{2} \ln(x^2 + 5) + C$$

Question 14 (4 marks) *[Definite Integral of exponential/logarithmic or trigonometric functions]*

Find the exact value of $\int_0^{\frac{\pi}{2}} (1 - \sin 2x) dx$.

$$\begin{aligned}\int_0^{\frac{\pi}{2}} (1 - \sin 2x) dx &= \left[x + \frac{\cos 2x}{2} \right]_0^{\frac{\pi}{2}} \\&= \left(\frac{\pi}{2} + \frac{\cos \pi}{2} \right) - \left(0 + \frac{\cos 0}{2} \right) \\&= \frac{\pi}{2} - \frac{1}{2} - \frac{1}{2} \\&= \frac{\pi}{2} - 1\end{aligned}$$

Question 15 (4 marks) *[Integration by Simple Substitution]*

Use substitution to evaluate $\int_2^3 \frac{x}{\sqrt{x^2 - 3}} dx$.

$$\text{let, } u = x^2 - 3$$

$$\frac{du}{dx} = 2x \Rightarrow du = 2x dx \Rightarrow \frac{du}{2} = x dx$$

$$\text{limits: } x = 2 \Rightarrow u = 2^2 - 3 = 1$$

$$x = 3 \Rightarrow u = 3^2 - 3 = 6$$

$$\begin{aligned}\int_2^3 \frac{x}{\sqrt{x^2 - 3}} dx &= \int_1^6 \frac{1}{\sqrt{u}} \frac{du}{2} \\&= \frac{1}{2} \int_1^6 u^{-\frac{1}{2}} du \\&= \frac{1}{2} \left[\frac{u^{\frac{1}{2}}}{\frac{1}{2}} \right]_1^6 = \left[u^{\frac{1}{2}} \right]_1^6 \\&= \sqrt{6} - \sqrt{1} = \sqrt{6} - 1\end{aligned}$$

Question 16 (3 marks) *[Integration by Simple Substitution]*

Find $\int \frac{\cos 2x}{2 \sin 2x} dx$ by substitution. Show all necessary working.

$$\text{let, } u = 2 \sin 2x$$

$$\frac{du}{dx} = 4 \sin 2x \Rightarrow du = 4 \cos 2x dx$$

$$\Rightarrow \frac{du}{4} = \cos 2x dx$$

$$\begin{aligned}\int \frac{\cos 2x}{2 \sin 2x} dx &= \int \frac{1}{2 \sin 2x} \cos 2x dx \\&= \int \frac{1}{u} \frac{du}{4} \\&= \frac{1}{4} \int \frac{1}{u} du \\&= \frac{1}{4} \ln |u| + c \\&= \frac{1}{4} \ln |2 \sin 2x| + c\end{aligned}$$

END OF EXAM